



Thực hiện Phân tích Tương quan bằng Excel

Sử dụng Ma trận tương quan
(Correlation Matrix) từ Data
Analysis Toolpak.

Mean	\$ 40,065.88
Standard Error	\$ 424.02
Median	\$ 28,276.00
Mode	\$ 3,269.00
Standard Deviation	\$ 44,043.13
Sample Variance	\$ 1,939,797,496.92
Kurtosis	\$ 8.19
Skewness	\$ 2.30
Range	\$ 468,597.00
Minimum	\$ 78.00
Maximum	\$ 468,675.00
Sum	\$ 432,270,823.00
Count	10,789.00

ILLUSTRATION 2.48 University Salary Data
Descriptive Statistics



Áp dụng (Apply It 2.4) - Kiểm toán Chi phí Bảo hành

Phân tích độ lệch chuẩn và histogram để phát hiện chi phí bảo hành bất thường.



2.5 Trực quan hóa Được Sử dụng Như thế nào?



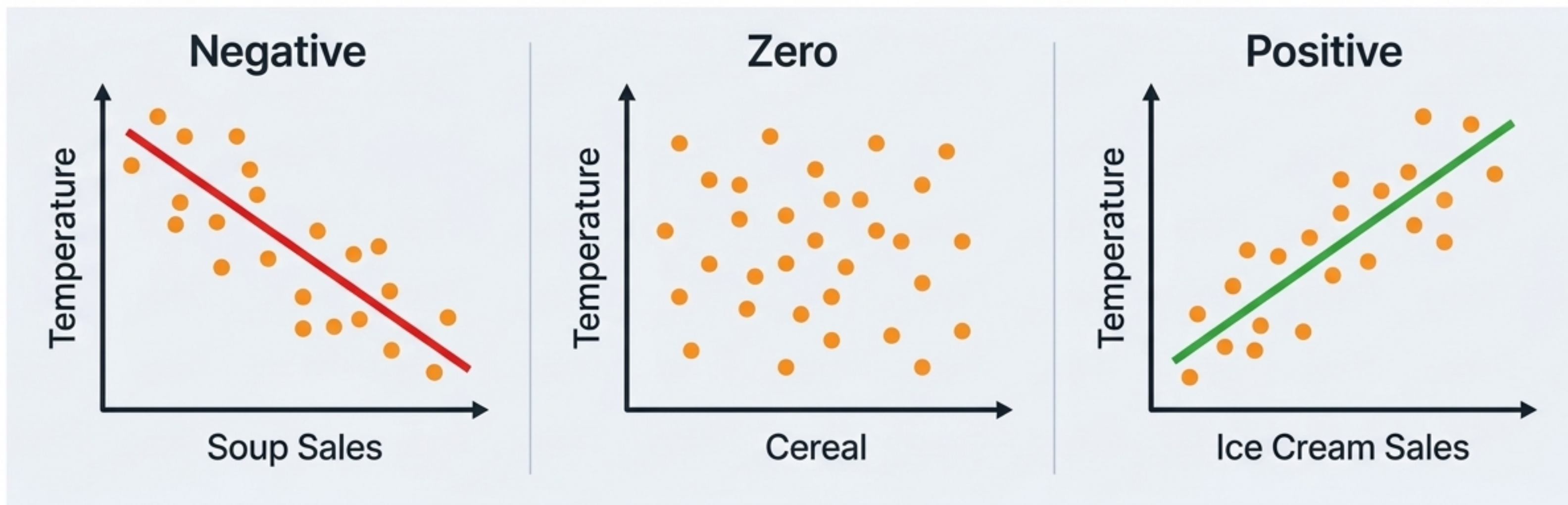
**60.000
lần**

Bộ não con người xử lý hình ảnh nhanh hơn **60.000** lần so với văn bản. Trực quan hóa giúp nhận diện các điểm dị biệt (outliers) và xu hướng (trends) không thể thấy trong các bảng số liệu.

“ Nguyên tắc: Bức tranh đáng giá ngàn con số



Các Dạng Tương quan (Phân tích Scatter)



Negative, Zero, Positive Scatter Plots

Trực quan hóa Hệ số Tương quan

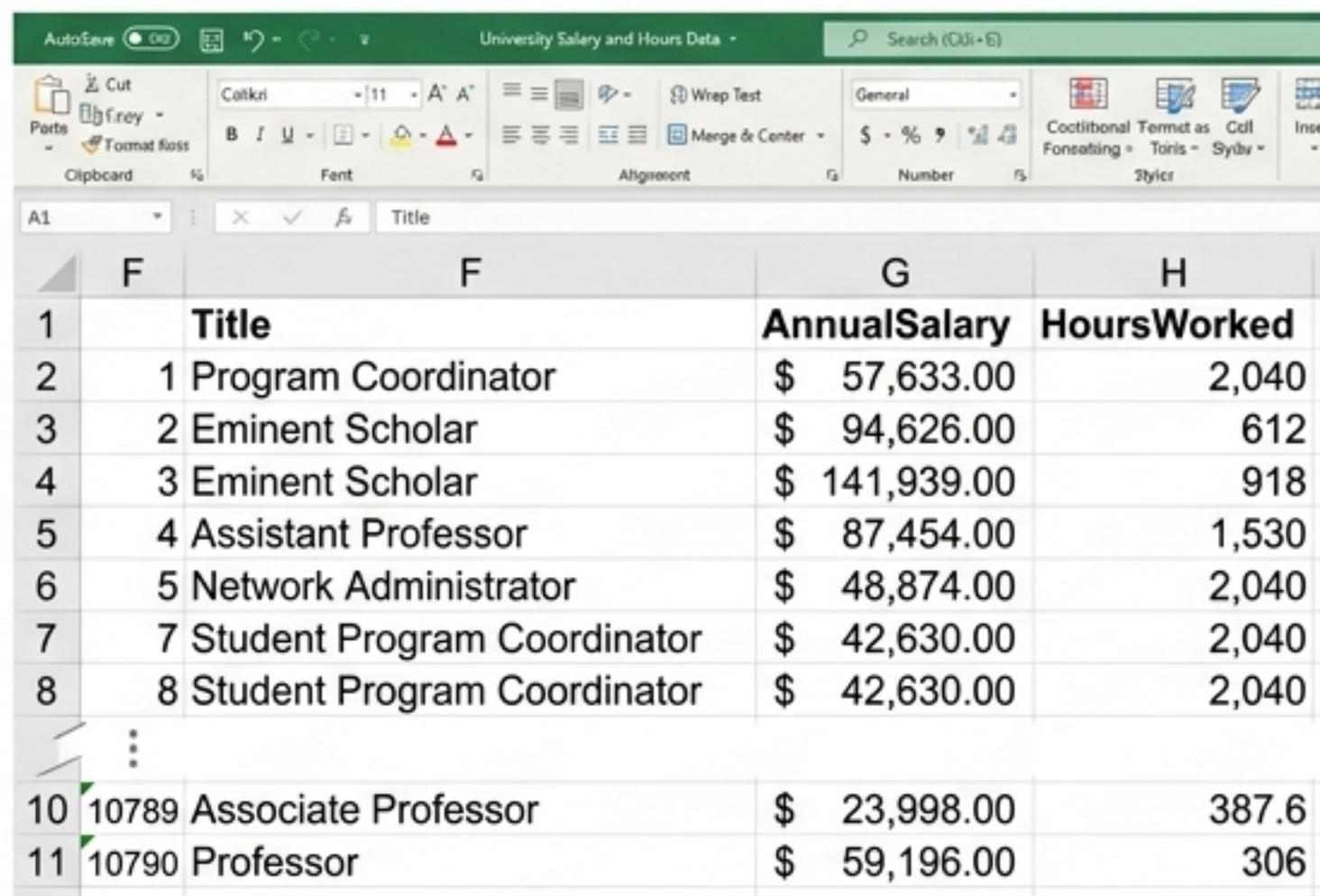


Correlation Coefficient of Ice Cream Sales and Temperature



Lợi ích của Trực quan hóa so với Bảng số liệu

Nhìn bảng số liệu khó phát hiện xu hướng.



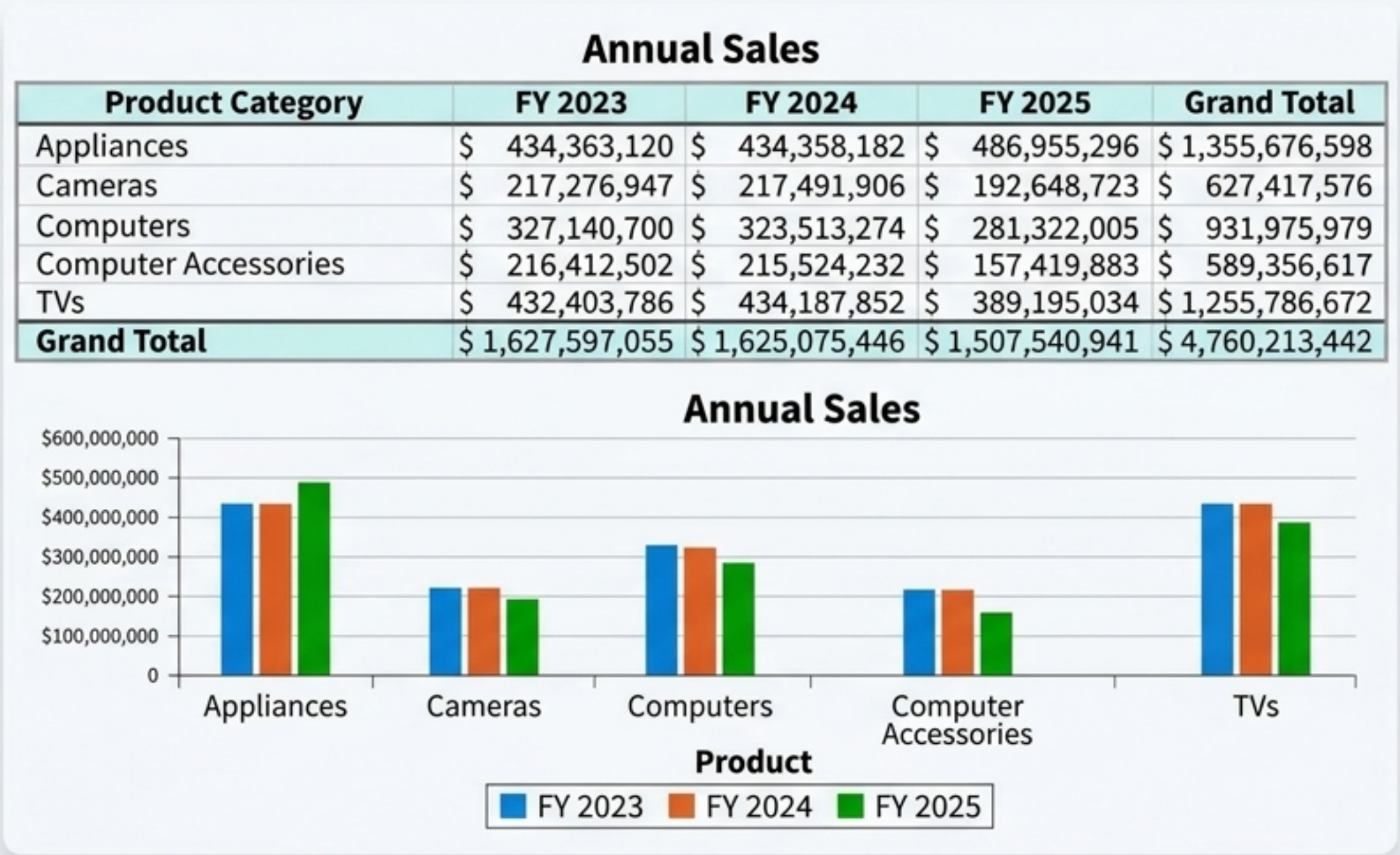
	F	F	G	H
1		Title	AnnualSalary	HoursWorked
2	1	Program Coordinator	\$ 57,633.00	2,040
3	2	Eminent Scholar	\$ 94,626.00	612
4	3	Eminent Scholar	\$ 141,939.00	918
5	4	Assistant Professor	\$ 87,454.00	1,530
6	5	Network Administrator	\$ 48,874.00	2,040
7	7	Student Program Coordinator	\$ 42,630.00	2,040
8	8	Student Program Coordinator	\$ 42,630.00	2,040
10	10789	Associate Professor	\$ 23,998.00	387.6
11	10790	Professor	\$ 59,196.00	306

ILLUSTRATION 2.53 University Salary and Hours Data



Giải pháp Trực quan: Biểu đồ

Biểu đồ đường (Line Chart) lập tức cho thấy sự sụt giảm doanh số ở một số vùng.



Annual Sales - Table and Chart



Lựa chọn Dạng Trực quan hóa (Decision Tree) Thực hành trên Excel

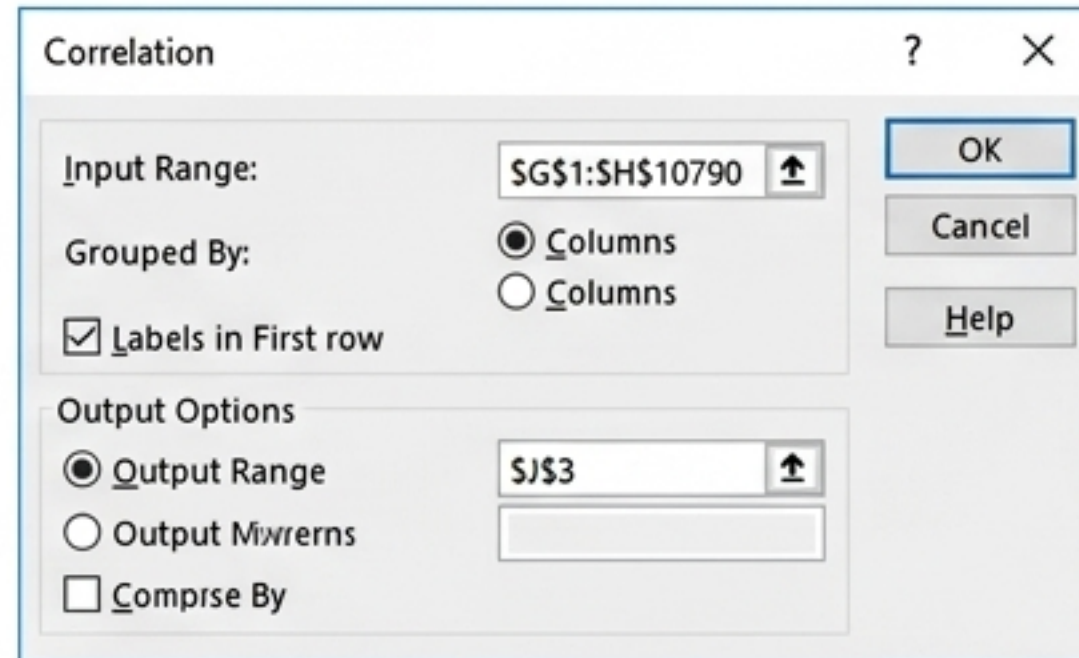
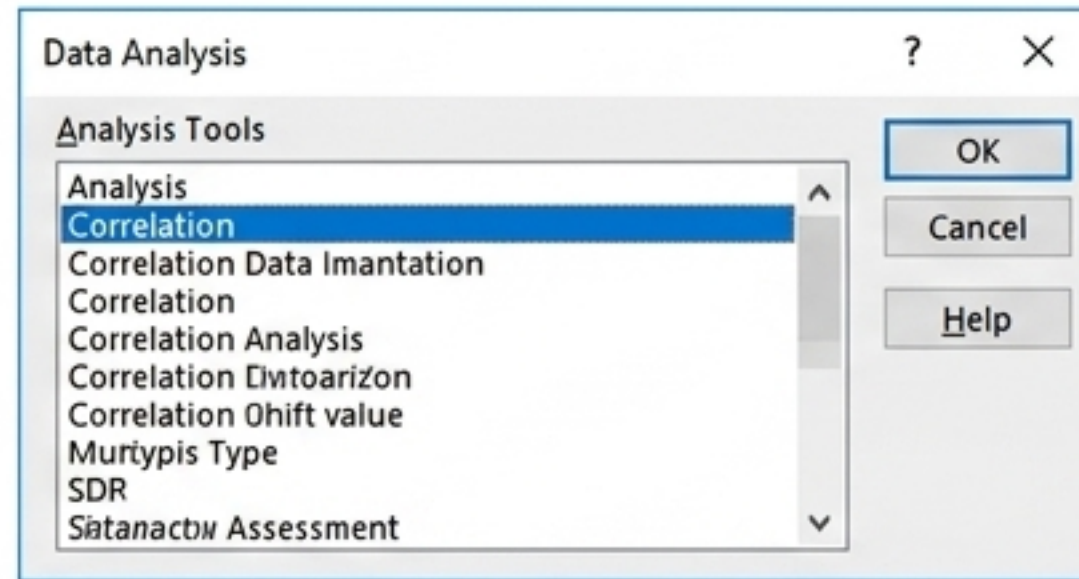


ILLUSTRATION 2.54 Data Analysis Tool and Correlation Dialog Box



Các Dạng Biểu đồ Phổ biến trong Excel (1/2)

Biểu đồ Cột (Column) & Thanh (Bar). Biểu đồ Phân tán (Scatter).

Visualization	Use	Best Practices	Data Required
Area Chart 	Representing changes in volume over time.	• Representing changes in volume over time. • Representing a organization. • Representing changes in volume over time. • Representing into practicalist and canstption.	Data required - data with group and dinancers
Bar and Column Chart 	Representing changes in nurkets of bar and column chart.	• Representing changes in bar and column chart. • Representing a organization. • Representing collaborat bar and column chart. • Representing columns in volume over time. • Representing into practicalist and canstption.	Data required - data with group and dinancers
Bubble Chart 	Representing changes in nurkets of b and bubble chart.	• Representing changes in bubble chart. • Representing a organization. • Representing remover focus for bubble charts. • Representing into practilcarist and cannerption.	Data required - data with exam and dinancers
Histogram Chart 	Representing changes in representing histogram volume over time.	• Representing changes in histogram in histogram chart. • Representing a organization. • Representing remover focus for in histograms. • Representing into practilcarisa and cannerption.	Data required - data with exam and dinancers



Cây quyết định để chọn biểu đồ: So sánh, Thành phần, Phân phối, hay Tương quan?

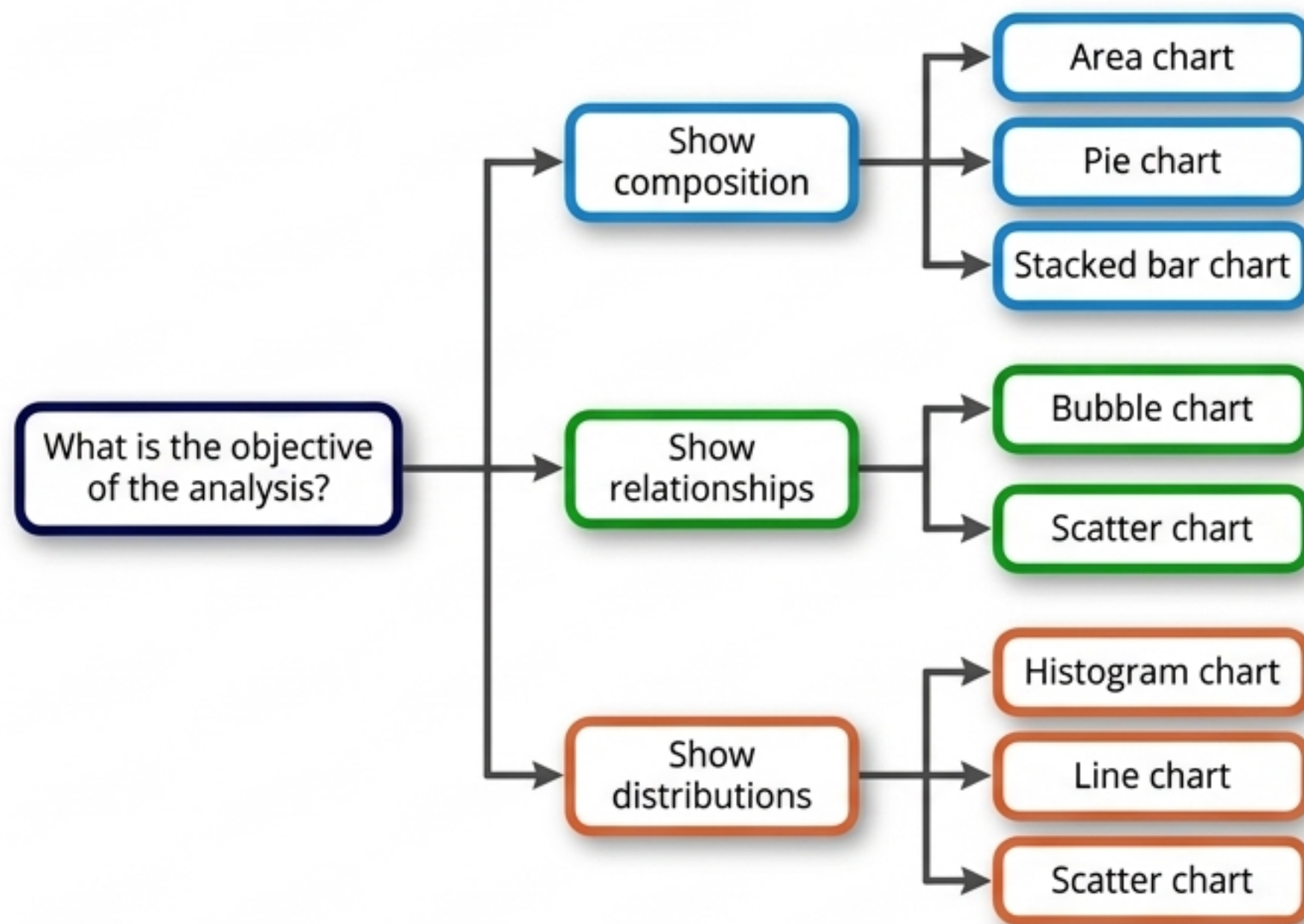


ILLUSTRATION 2.58 Guide to Visualizations that Show Composition, Relationships, and Distributions



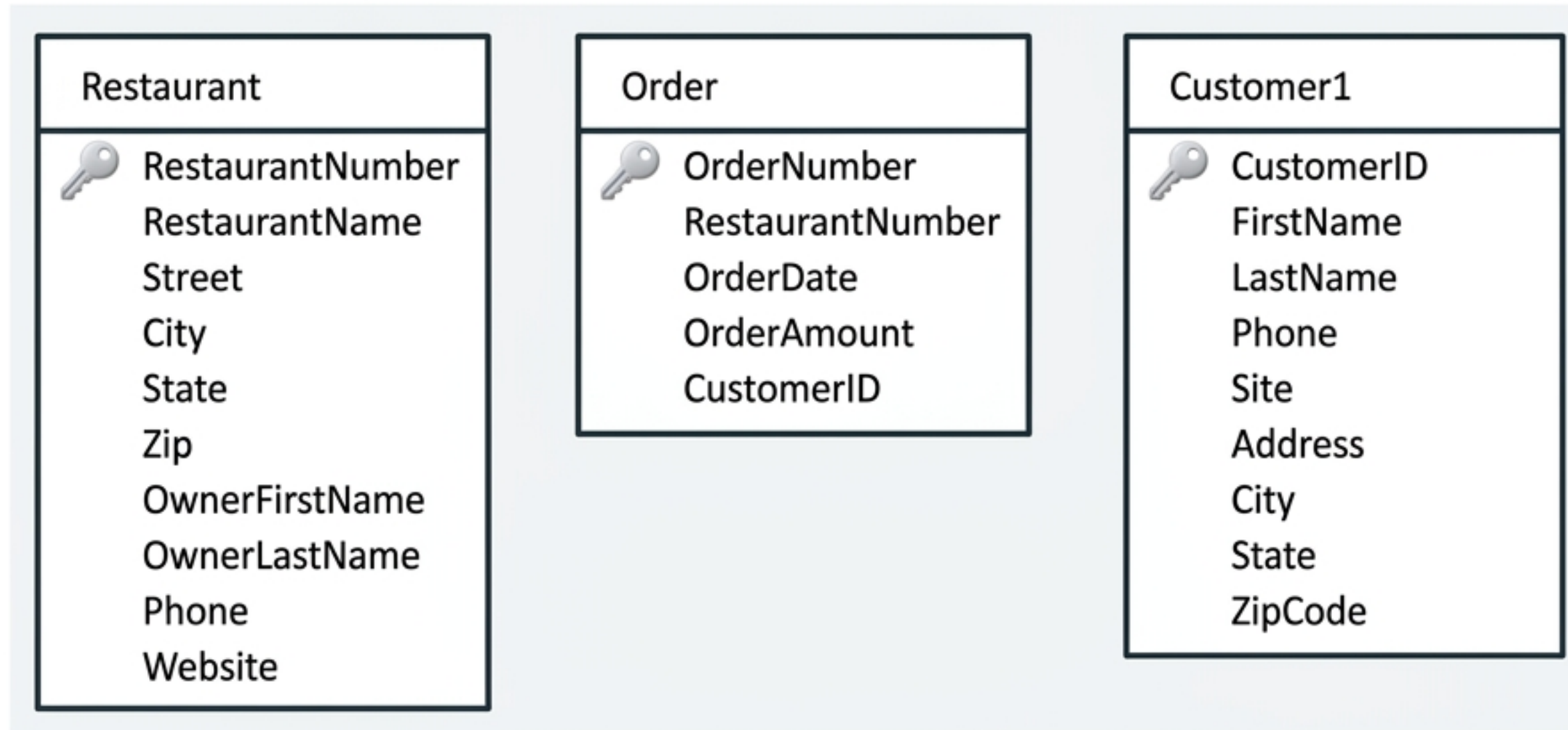
Đánh giá và Thực hành Chương 2

Phần Bài tập Thực hành (Practice & Exercises)

TakeOrder	Customers	Employees
- OrderNumber - CustomerID - EmployeeID - Date - Time - Notes	- CustomerID - LastName - FirstName - StreetAddress - City - State - ZipCode - PhoneNumber	- EmployeeID - LastName - FirstName - DateHired - DateofBirth - PayRate(hourly) - JobTitle - StreetAddress - City - State - ZipCode - PhoneNumber

TakeOrder, Customers, Employees Database Schema

Bài tập Lược đồ Cơ sở Dữ liệu (1)



Restaurant, Order, Customer1 Database Schema

Bài tập Ngắn (Brief Exercises) - Hình: BE 2.1 & Hình: BE 2.2



Locations LocationNumber LocationDescription	SalesOrders SalesOrderNumber OrderNumber Model SoldDate SalesVolume Color Location Region State Country UnitSalePrice CustomerNumber EmployeeNumber	Employee EmployeeNumber FirstName LastName Address City State ZipCode RegionNumber
Regions RegionNumber RegionDescription		
Inventory ItemNumber ItemDescription Color QuantityOnHand	SalesOrderExpenses OrderNumber VariableMarketing Labor Materials Overhead TotalWarranty TotalDepreciation SalesTax	Customer CustomerNumber CustomerName CustomerAddress CustomerCity CustomerState CustomerZipCode ContactFirstName ContactLastName PhoneNumber

Hình: BE 2.1

Locations LocationNumber LocationDescription	SalesOrders SalesOrderNumber OrderNumber Model SoldDate SalesVolume Color Location Region State Country UnitSalePrice CustomerNumber EmployeeNumber	Employee EmployeeNumber FirstName LastName Address City State ZipCode RegionNumber
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Inventory ItemNumber ItemDescription Color QuantityOnHand	SalesOrderExpenses OrderNumber VariableMarketing Labor Materials Overhead TotalWarranty TotalDepreciation SalesTax	Customer CustomerNumber CustomerName CustomerAddress CustomerCity CustomerState CustomerZipCode ContactFirstName ContactLastName PhoneNumber

Hình: BE 2.2

Bài tập (Exercises - EX 2.1 & EX 2.2)

<table><tr><th>Locations</th></tr><tr><td> LocationNumber LocationDescription</td></tr></table>	Locations	 LocationNumber LocationDescription	<table><tr><th>SalesOrders</th></tr><tr><td> SalesOrderNumber OrderNumber Model SoldDate SalesVolume Color Location RegionNumber State Country UnitSalePrice CustomerNumber EmployeeNumber</td></tr></table>	SalesOrders	 SalesOrderNumber OrderNumber Model SoldDate SalesVolume Color Location RegionNumber State Country UnitSalePrice CustomerNumber EmployeeNumber	<table><tr><th>Employee</th></tr><tr><td> EmployeeNumber FirstName LastName Address City State ZipCode RegionNumber</td></tr></table>	Employee	 EmployeeNumber FirstName LastName Address City State ZipCode RegionNumber	<table><tr><th>Locations</th></tr><tr><td> LocationNumber LocationDescription</td></tr></table>	Locations	 LocationNumber LocationDescription	<table><tr><th>SalesOrders</th></tr><tr><td> SalesOrderNumber OrderNumber Model SoldDate SalesVolume Color Location Region State Country UnitSalePrice CustomerNumber EmployeeNumber</td></tr></table>	SalesOrders	 SalesOrderNumber OrderNumber Model SoldDate SalesVolume Color Location Region State Country UnitSalePrice CustomerNumber EmployeeNumber	<table><tr><th>Employee</th></tr><tr><td> EmployeeNumber FirstName LastName Address City State ZipCode RegionNumber</td></tr></table>	Employee	 EmployeeNumber FirstName LastName Address City State ZipCode RegionNumber
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Bài tập (Exercises - EX 2.3 & EX 2.4)

Tình huống Ứng dụng Chuyên môn (PAC) - Pizza My Heart Food Truck

Tables		Fields
Tables	Sales and Customer	1. SalesOrderID, CustomerID, OrderDate, TotalAmount
		2. CustomerID, FirstName, LastName, Address, City, State, ZipCode, PhoneNumber
		3. _____
		4. _____
		7. _____
		8. _____

Dữ liệu Tình huống & Dữ liệu Tình huống (Tiếp)

Question	Table	Fields	Join type
9. Are there vendors that have not made purchases?			
10. Do any employee addresses match vendor addresses?			
11. What is the total sales by menu item?			
12. What are the total sales by truck?			
13. Are there purchases made from vendors that are not in the Vendor table?			



Kết thúc. Hỏi & Đáp. Cảm ơn các bạn đã lắng nghe!